

Introduction Roughs for SPECTRUM

Drafted 2026-09-13; finalized after three rounds of adversarial review

This file contains seven candidate Introductions for `01_slr_forecast_intervention2026.tex`. Version 0 was written last, in a separate session, as a short introduction that carries the essential material and leaves the supporting detail of Versions 1–6 to the body of the paper. It was subsequently revised for the audience of the AGU Chapman Conference on sea-level rise projection, which raised its citation density and its technical specificity. A later pass cut most of the West Antarctic discussion, on the grounds that this study represents that uncertainty through a scenario mixture rather than resolving it, and reinvested the space in the central argument, which is the value of a hierarchy of forecast models that can each be calibrated and validated against data. A further pass recast what remained of the process-model discussion in general terms. It replaced the catalogue of Antarctic process detail with the field-wide evidence that these models are underconstrained relative to the observational record, hindcast poorly, carry ensemble spreads that are not reliable uncertainty estimates, depend on initialization, and depend on calibration choices as much as on physics. That pass also replaced the non sequitur that followed the AR6 projection range, which had pivoted without warning into ISMIP6, LARMIP-2, and structured expert judgment, with a statement of how wide the published high-end range now is and of how the underlying ice-sheet models perform against the historical record. West Antarctica is now named only once, in the closing paragraph, and only to explain why that contributor is represented as a scenario mixture. The general evidence retained in the model-class paragraph is drawn in part from Antarctic studies but is framed as evidence about sea-level projection as a whole. That pass dropped ten keys (Bamber2019, Edwards2019, Edwards2021, Gudmundsson2013, HaseloffSergienko2018, Levermann2020, Martin2026, Pegler2018, Ranganathan2024, and Seroussi2020) and added none. A final pass supplied references for the hierarchy-of-models paragraph, which had been an uncited block of argument and would not have been publishable as one. It now carries Held2005, Jeevanjee2017, and Maher2019 for the model-hierarchy practice itself, Oreskes1994 for the point that a model of a natural system cannot be verified against truth, Aschwanden2021 and Felikson2025 for the formal role of observations in refining projections, Lipscomb2025 for the weight carried by agreement among models that do not share assumptions, and Green2015 with Makridakis2020 for the value of simplicity, whose published definition includes being understandable by those who must act on the forecast. Version 0 now runs to 85 body lines, or 1.85 pages as typeset at 46 lines per page, and cites 48 keys. Of those, Lipscomb2025 was verified against a local Zotero PDF and added to `references.bib`, and Jeevanjee2017, Maher2019, and Oreskes1994 were added as new entries reconstructed from recall rather than from a publisher record or a local PDF. Those three are flagged

in `references.bib` and every field in them must be confirmed before submission, as must Held2005, which was already present and carries the same flag. Every other key is either already used in the manuscript itself or in Versions 1–6 of this file, or was verified against a local Zotero PDF (Goldberg2026 and Morlighem2024). Versions 1–6 were produced as directed: Version 1 follows the six-item bulleted outline already present at the top of that manuscript’s `\section{Introduction}`; Version 2 is an independently constructed introduction drawing on the broader survey in `literature_component_slr.tex` and the project’s own findings; Version 3 is a condensed synthesis of the two, retaining what we judged the most load-bearing points (the stakes, the three-way model-class trade-off, the WAIS tail-risk gap, and the SPECTRUM framework itself) while dropping supporting detail and secondary citations; Versions 4 and 5 are two independent critical edits of Version 3 (described further below) that substantially expand its treatment of West Antarctic Ice Sheet (WAIS) uncertainty and restructure its sentences for classic-style, low-embedding prose; Version 6 merges Versions 3–5 into a single text and corrects a logic error common to all three (described further below). All six are written to match the voice, quantitative style, and citation density of the target manuscript. Versions 1 and 2 were each drafted, then put through three rounds of independent adversarial review (fresh reviewers each round, checking citation validity against `references.bib`, accuracy against the papers actually cited, style conformance, and prose quality) with a full revision pass after each round, and were confirmed in the final round to compile cleanly (verified with a live `pdflatex/bibtex` run) with no outstanding material issues. Version 3 draws only on claims and citations already verified in Versions 1 and 2, and received one adversarial review pass of its own (citation validity, accuracy, and length target) rather than the full three rounds, followed by a second revision pass (described below) and, in Versions 4 and 5, a third round that substantially expands the WAIS discussion with newly sourced, independently verified citations. Nothing here has been copied into the manuscript itself.

Length note. Version 1 is 1044 words and Version 2 is 872 words (body text only, excluding headings, counted programmatically). Version 3 originally targeted the intersection of “half to two-thirds” of each, i.e. 522–581 words, and its first draft came in at 580 words, inside that band relative to both ($0.56 \times V1$, $0.67 \times V2$). A second revision pass (described below) added connective sentences to fix structural and clarity problems surfaced by adversarial review; the current text is 623 words ($0.60 \times V1$, $0.71 \times V2$), modestly above the original band, a trade-off made deliberately in favor of clarity over the length target. Its citation count (23) is unchanged by that pass and was scaled down roughly in proportion to its length rather than held to the 25–40 range specified for Versions 1 and 2, since forcing that many citations into a passage this short would come at the expense of prose quality. Version 3’s first draft received one full adversarial review pass, which caught (and that pass fixed) several compression artifacts: a fabricated causal link between the model-parsimony citations and the held-out-validation design, an inaccurate collapsing of Rahmstorf (2007) and Vermeer & Rahmstorf (2009) into a single-coefficient description (the latter has two), a dropped hierarchy-of-models rationale, an orphaned citation, and a reversed description of what component-level models trade away.

Revision note (second pass). The revised draft was then checked by three independent adversarial reviewers, each rereading the full text and mapping every sentence to the idea it is meant to convey, one focused on argumentative structure and logical coherence, one

on plain-language and sentence-level clarity, and one on whether each sentence’s content actually matched the goal that mapping assigned it. All three independently flagged the same unresolved referent (the phrase “that gap” opening Paragraph 3, which pointed to nothing named in the preceding paragraph); two flagged a component-level-models sentence that conflated a stated weakness (assuming stationarity) with an unrelated, more important claim (no published implementation withholds the aggregate record for an independent check) behind a single semicolon; and reviewers separately flagged an unearned closing claim in Paragraph 2 (that ordering model classes by complexity lets simpler ones check more complex ones, asserted without the paragraph showing how), an unsupported closing claim in Paragraph 3 (an appeal to “adaptation planning” introduced with no groundwork), and several sentences overloaded with two or more distinct claims joined by colons or semicolons. The text above fixes all of these: it names the referent explicitly (“this missing independent check”), splits the conflated and overloaded sentences, ties the Paragraph 2 closing claim to the specific held-out GMSLR–GMST check performed in Paragraph 4, and rewrites the Paragraph 3 closing claim to point back to the flooding-damage stakes already established in Paragraph 1 rather than to an unintroduced concept. No citations were added or removed in this pass.

Revision note (third pass: Versions 4 and 5). Two independent editors were each given the post-second-pass text of Version 3 and the same brief: cut any sentence not earning its place, substantially expand the WAIS paragraph with additional citations supporting the claim that WAIS is the largest single source of sea-level-projection uncertainty (not limited to a pre-supplied citation pool; both were explicitly told to search the web for further real, verifiable literature), and restructure sentences per the classic-style and syntax-tree principles in Steven Pinker’s *The Sense of Style* (subject and verb kept close together, complexity pushed to the edges of a sentence rather than center-embedded, end-weighting, avoiding nominalizations, one claim per main clause). Working independently, one editor (Version 4) built a thesis-first, explicitly numbered four-line evidentiary case (observational trend, independent expert judgment, multi-model spread, and a temporal-variance decomposition), closing with a separate mechanism paragraph; the other (Version 5) built a three-movement prose argument (mechanism, observation, convergence) that additionally surfaces a genuine counter-finding (*Edwards et al.* 2019 finds ice-cliff collapse is not required to match the observational record, bounding the 95th-percentile Antarctic contribution below 43 cm without it) and argues this makes West Antarctic uncertainty structural rather than narrowing it. Every new citation both editors introduced was checked against a primary source (the publisher page, a DOI resolver, or the Crossref API) before use, not recalled from training data; four candidate citations were considered and rejected as redundant with citations already used. The coordinating session independently re-verified the four most load-bearing new numeric claims (the *DeConto and Pollard* 2016 “more than a meter by 2100” figure, the *Edwards et al.* 2019 43 cm bound, the *Seroussi et al.* 2020 -7.8 to $+30.0$ cm ISMIP6 range, and the *Milillo et al.* 2019 tenfold spatial variability in Thwaites melt rates) against their original abstracts via web search and found all four accurate as stated in both drafts.

Revision note (fourth pass: Version 6). Versions 3, 4, and 5 all opened the WAIS discussion with some variant of “the stakes of [the missing independent check] are concentrated in West Antarctica.” This was flagged as a logic error: it frames West Antarctica’s dominance as a consequence of, or contingent on, the field-wide calibration gap identified

two paragraphs earlier, when in fact West Antarctica would be the largest source of sea-level-projection uncertainty regardless of whether any model class withholds an independent validation target. The two facts are real but independent, and stringing them together with “the stakes of X are concentrated in Y” asserts a causal or scoping relationship between them that the surrounding text never establishes. Version 6 merges the strongest elements of Versions 3–5 (Version 4’s thesis-first framing and tighter model-class paragraph; Version 5’s Edwards2019 counter-finding, Milillo2019 and Levermann2020 citations, and punchier closing sentences; the full citation union of both WAIS expansions, with none dropped) and repairs the transition by (i) stating West Antarctica’s dominance as a freestanding claim, explicitly denying that it follows from the calibration gap (“That is not a consequence of how any model class chooses to calibrate; it follows from the ice physics and the observational record themselves”), and (ii) moving the correct, oppositely-directed causal claim to the start of the SPECTRUM paragraph, where it belongs: the calibration gap matters *because* West Antarctica’s stakes are largest, not the reverse (“the gap matters most exactly where the stakes are largest: West Antarctica”). No citations were added beyond the union already present in Versions 4 and 5, and none were dropped.

Bibliography note. Version 1 cites only keys that already existed in `references.bib` at the time it was drafted. Version 2 additionally cited several papers discussed in `literature_component_slr.tex`: Felikson2025, Slangen2022, Garner2018, Nauels2017, Kopp2023, McCormack2026, Bakker2017, Darnell2025, and Held2005; these have since been added to `references.bib` (by an earlier session, prior to the WAIS-expansion pass described above), so all citations in Versions 1–3 now resolve. Versions 4 and 5 introduce four further keys, added to `references.bib` as part of this pass: Kopp2017, Bulthuis2019, Milillo2019, and Levermann2020. Like McCormack2026 and Held2005 before them, these four are flagged in `references.bib` with a note that they were confirmed against the publisher record or the Crossref API rather than a local Zotero PDF, and should be checked against a local PDF before submission, per this project’s citation-verification standard. Both new sections also reuse the existing DeContoP2016 and Edwards2019 keys (already present in `references.bib` and already cited in Version 1) rather than introducing duplicate entries for the same two papers.

Bibliography note (fifth pass: Nerem2022). A fifth key, Nerem2022 (*Nerem et al.* 2022: a purely empirical quadratic extrapolation of the satellite-altimetry GMSL record itself, with no temperature regression, found to agree with IPCC SROCC and AR6 21st-century projections), was added to `references.bib` from a local Zotero PDF and cited in all six versions above, in each version’s semi-empirical paragraph, as a still simpler empirical benchmark alongside Rahmstorf2007/Vermeer2009: unlike those, it needs no GMST regression at all, extrapolating the observed sea-level record directly. No existing sentences were otherwise altered.

Version 0 — short introduction written to the manuscript’s voice

Every centimeter of that rise exposes roughly 1.6 million more people to annual coastal flooding (*Kulp and Strauss*, 2019), and without further adaptation annual flood damages reach US\$14–18 trillion per year at 0.9–1.1 m of rise (*Jevrejeva et al.*, 2018). What can be

done about that depends on how skillfully we can predict the rate of sea-level rise. Forecasts published over four decades span a wide range of methods and outcomes, and the high-end estimates published since the Sixth Assessment Report of the Intergovernmental Panel on Climate Change (AR6) span 0.6–2.0 m of rise by 2100 (*Garner et al.*, 2026). Coastal planning leans on AR6 itself, which projects 0.4–0.7 m this century (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). The ice-sheet models underlying those land-ice projections reproduce the historical record poorly. Few match observed mass loss, and the spread across an ensemble of model outputs is not a reliable measure of the uncertainty it is taken to represent (*Aschwanden et al.*, 2021).

Global warming that has happened and will happen ensure that projections of sea-level rise for the 21st century and beyond are out of sample with respect to the modern observational record no matter the projection method. The central difficulty is therefore not building a model but calibrating it, testing it against data withheld from that calibration, and reporting uncertainties in a form that separates the contribution of each physical process. A recent community synthesis makes this case, calling for projections narrowed by observations and for uncertainty characterized by its structure and not only by its magnitude (*Feliksion et al.*, 2025).

Sea-level projection models span a wide range of complexity, and their strengths are complementary. Process-based ice-sheet models solve the equations of ice flow, deformation, and fracture, and they are where competing hypotheses about the mechanisms of ice loss can be posed against one another and tested (*Schoof*, 2007; *Morlighem et al.*, 2024). Their parameters, such as the basal sliding coefficient, are inferred from observations by ill-posed inverse problems in which the data constrain only a limited set of parameter directions (*Isaac et al.*, 2015; *Petra et al.*, 2014). They depart from reconstructions of historical GMSLR in hindcast (*Aschwanden et al.*, 2021), and most projections initialized in 2016 grow the Antarctic Ice Sheet through the years that overlap the satellite record, with outcomes that remain sensitive to the state a model starts from (*Fricker et al.*, 2025). Structural assumptions carry comparable weight. The stress exponent of the flow law for ice is usually assigned a value lower than observations of fast-flowing ice shelves indicate (*Millstein et al.*, 2022), and where grounding lines retreat rapidly, uncertainty in that exponent produces a wider spread in projected ice loss than uncertainty in climate forcing does (*Getraer and Morlighem*, 2025). Calibration can hide such a bias, because a model that assumes $n = 3$ can be initialized to match observations of an $n = 4$ glacier yet underestimate its 100-year sea-level contribution by 21 to 35% in retreat benchmarks (*Martin et al.*, 2026). Two ice-sheet models calibrated against different observations of the same glacier give mid-century loss rates differing by more than a factor of two (*Goldberg et al.*, 2026).

At the other extreme, extrapolations of the observed record and semi-empirical fits of the rate of GMSLR to global mean surface temperature (GMST) are transparent and anchored in the quantity being predicted (*Nerem et al.*, 2022; *Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Kopp et al.*, 2016), and they track the observed record at least as well as contemporaneous process-based projections (*Rahmstorf et al.*, 2012). They give up any account of which processes are responsible, and the fitted relationship need not hold as individual contributors respond on different timescales (*Church et al.*, 2013). Between the two sit component-level models, which separate GMSLR into its physical contributors and calibrate each one against observations, model output, or expert assessment (*Kopp et al.*,

2014; Mengel *et al.*, 2016; Wong *et al.*, 2017; Perrette *et al.*, 2013; Nauels *et al.*, 2017; Kopp *et al.*, 2023), trading spatial detail for parsimony and a direct link to data.

A numerical model of a natural system can be tested against data and against other models, but it cannot be shown to be correct (Oreskes *et al.*, 1994). No one class is at once the most physically detailed, the most transparent, and the most tightly constrained by data, so treating any one of them as authoritative discards what the others offer. Ordered by complexity they instead form a hierarchy, a practice long established elsewhere in climate science, where ordering models from idealized to comprehensive has served understanding better than comprehensive models alone (Held, 2005; Jeevanjee *et al.*, 2017; Maher *et al.*, 2019). Each level can be calibrated and tested against data before the next is trusted, giving observations the formal role in refining projections that recent assessments of ice-sheet and sea-level projection have called for (Aschwanden *et al.*, 2021; Felikson *et al.*, 2025). An extrapolation of the observed record sets a scenario-independent reference that any more elaborate model should be able to reproduce. A semi-empirical fit adds a dependence on warming and little else, which isolates that sensitivity and makes it testable on its own. A component-level model then asks whether the same sensitivity can be recovered from the separate physics of the ocean, the glaciers, and the ice sheets, each calibrated against its own record. A process-based model asks which mechanisms within those contributors are responsible. Calibration and validation are then carried out within the space of models rather than against a single preferred one, and agreement between levels becomes a diagnostic in its own right. Where the levels agree, a result rests on more than one set of assumptions, and agreement of that kind carries more weight than agreement among models that share their assumptions (Lipscomb *et al.*, 2025). Where they diverge, the divergence localizes what the data do not yet constrain and identifies the measurement that would settle it. The simplest levels often carry the most transferable insight, because what they assume is small enough to state plainly, to test directly, and to be understood by those who must act on it (Green and Armstrong, 2015; Makridakis *et al.*, 2020). A hierarchy built this way complements process-based modeling rather than substituting for it.

Here we construct a data-driven, component-level forecast of 21st-century sea-level rise. Where the observational record can constrain a contributor to GMSLR, we give it a parsimonious physical model calibrated by Bayesian inference against an independent record spanning decades, among them NOAA thermosteric estimates (Levitus *et al.*, 2012), EN4 subsurface ocean temperature (Good *et al.*, 2013), Greenland outlet-glacier discharge (Mouginot *et al.*, 2019), GlaMBIE (The GlaMBIE Team *et al.*, 2025), and IMBIE-3 (Otosaka *et al.*, 2026). The remaining contributors, among them Greenland and East Antarctic surface mass balance and terrestrial water storage, are adopted from published model sensitivities and assessments (Fettweis *et al.*, 2020; Noël *et al.*, 2021; Frieler *et al.*, 2015; Frederikse *et al.*, 2020). We call the framework SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Most of its component models are adapted from formulations in the literature, and for those our contribution lies in the calibration data, the validation, and the treatment of uncertainty. The exception is Greenland Ice Sheet discharge, for which we develop a three-parameter model driven by the temperature of the surrounding ocean. West Antarctic discharge is the one contributor we do not fit to warming, because the observational record cannot yet distinguish a self-sustaining retreat from one sustained by ocean melting from below (Sergienko *et al.*, 2026; Hill *et al.*, 2023; Reese

et al., 2023), and we represent it instead as a structured mixture of scenarios rather than a single trajectory. We do not attempt to settle that question here. Observed GMSLR and its sensitivity to observed GMST are withheld from every component calibration and reserved as diagnostics for the summed model, so the central test of the framework is out of sample. The independently calibrated components close the satellite altimetry budget (*Beckley et al.*, 2017; *Hamlington et al.*, 2020) and reproduce the level, the rate, and the transient sensitivity of GMSLR since 1950 across roughly 1 °C of warming (*Frederikse et al.*, 2020; *Fox-Kemper et al.*, 2021), a record none of them ever saw.

Version 1 — follows the manuscript’s own itemized outline

Coastal communities plan against a moving target. Each additional centimeter of global mean sea-level rise (GMSLR) exposes roughly 1.6 million more people to annual flooding (*Kulp and Strauss*, 2019), nearly 90% of them in low- and middle-income countries (*Jongman et al.*, 2012). Without further adaptation, annual flood damages are projected to reach US\$14–18T (trillion) per year once GMSLR reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018), while a lower-bound estimate of the cost of adapting to that rise, based on the price of raising dikes along a small fraction of the world’s coastlines, is on the order of US\$4T per meter (*Tiggeloven et al.*, 2020). Developing countries currently receive roughly US\$21B per year in adaptation finance across all sectors (*United Nations Environment Programme*, 2023), far short of what these cost estimates imply is needed. That shortfall makes the reliability of the sea-level projections used to plan adaptation directly consequential for how effectively it can be closed.

Sea-level projections have been published regularly for more than four decades and span a wide range of methods and outcomes (*Garner et al.*, 2026). Rates of GMSLR have already doubled twice since the mid-20th century, first over about 16 years and 0.34 °C of warming and again over about 21 years and 0.41 °C (*Frederikse et al.*, 2020; *Hamlington et al.*, 2024; *Rohde and Hausfather*, 2020). Yet current coastal planning still leans heavily on projections from the IPCC Sixth Assessment Report (AR6), which places 21st-century rise at 0.4–0.7 m (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). This variety is not a weakness so much as an underused resource: no single class of model is simultaneously the most physically detailed, the most transparent, and the most tightly constrained by data, and a framework that arrays these approaches by complexity, testing each level against data before trusting the next, offers a way to combine their respective strengths and to identify where projections agree, diverge, or exceed what the data can support (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020).

At the most detailed end sit numerical ice-sheet models, which resolve ice flow, deformation, and fracture directly and let competing physical hypotheses, such as whether marine ice-cliff instability can activate this century, be posed and debated within a common numerical framework (*DeConto and Pollard*, 2016; *Edwards et al.*, 2019; *Morlighem et al.*, 2024). Their principal weakness is that this physical detail demands more constraints than the observational record can currently supply. In hindcasting studies, process-based models diverge from historical GMSLR reconstructions (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). These disagreements trace to

structural gaps such as the omission of leading-order calving physics (*Fricker et al.*, 2025) and an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025), compounded by initialization uncertainty and, for marine ice sheets specifically, dynamics that can skew the resulting projections toward high outcomes rather than merely widening them (*Robel et al.*, 2019). The resulting parameterizations are numerous enough, and their interactions complex enough, that these models function largely as black boxes even to their developers, and every century-scale projection necessarily extrapolates them well beyond the short observational record used to tune them. Decades of general forecasting research show that models with more free parameters than their calibration data can constrain tend to forecast worse, not better, than simpler alternatives (*Green and Armstrong*, 2015).

At the opposite extreme are semi-empirical models that regress the observed rate of GM-SLR directly onto global mean surface temperature (GMST), a class whose transparency and direct grounding in the data being predicted make it easy to test and to communicate (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009). Statistically, semi-empirical fits track the observed sea-level record over multi-decade hindcast windows at least as well as, and at times better than, contemporaneous process-based projections (*Rahmstorf et al.*, 2012), and Bayesian treatments of the same GMST-GMSL relationship, which propagate measurement and proxy uncertainty explicitly rather than fitting a single point estimate, extend the approach over far longer records (*Kopp et al.*, 2016). Their central limitation is that a single global regression coefficient is not obviously physical or causal, may not stay stationary as individual components respond to different forcings on different timescales, and gives no way to tell, when it forecasts correctly, whether that agreement reflects the right physical reasons (*Church et al.*, 2013). An even simpler benchmark needs no temperature regression at all: extrapolating a quadratic fit to the satellite-observed GMSL record itself agrees with IPCC SROCC and AR6 projections to within their own stated uncertainty (*Nerem et al.*, 2022).

Between these extremes sit component-level models that decompose GMSLR into its physical contributors, ocean thermal expansion, glaciers, and the ice sheets, and calibrate each against existing observational or model data (*Kopp et al.*, 2014; *Mengel et al.*, 2016; *Wong et al.*, 2017). This intermediate complexity is a deliberate trade. Parsimony and a direct connection to physical processes come at the cost of lumped parameters that cannot resolve individual basins or glaciers, an assumption of stationarity over the calibration period, and the same extrapolation risk that limits every other class of model. The calibration strategy within this class still varies: some scale a component decomposition to modeled temperature trajectories (*Perrette et al.*, 2013), others use Bayesian-calibrated reduced-complexity models, with an expert prior on the most uncertain term (Antarctic dynamics), to bound a physically plausible upper limit on total sea-level rise (*Sriver et al.*, 2012), and still others forgo new component physics altogether and instead statistically fuse existing probabilistic projections into a single consensus estimate (*Grandey et al.*, 2024).

A framework that spans this hierarchy, from semi-empirical regression through component-level, data-calibrated models to full process-based simulation, does not need to choose a single class as authoritative. It can use agreement or disagreement across levels as a diagnostic in its own right, letting the simplest, most transparent models motivate and check the more complex ones, calibrating and validating within the space of models rather than against a single preferred one, and it can let the most consequential insights come from the simplest

level rather than the most complex. Here, we construct such a framework: a data-driven, component-level sea-level projection that treats each contributor to GMSLR, ocean thermal expansion, Greenland Ice Sheet (GrIS) surface mass balance (SMB) and glacier discharge, West Antarctic Ice Sheet (WAIS) glacier discharge, East Antarctic Ice Sheet (EAIS) SMB, the Antarctic Peninsula, glaciers outside the ice sheets, and terrestrial water storage, with its own parsimonious physical model, calibrated against independent observational records using Bayesian inference. We call this framework SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Most of its component models adapt existing formulations from the literature; the exception is Greenland Ice Sheet discharge, whose 3-parameter model connecting glacier discharge directly to ocean temperature is developed here. Critically, observed GMSLR and its sensitivity to observed GMST are withheld from every component-level calibration and used only as diagnostics on the composite model, so that the framework’s central test, whether independently calibrated parts sum to reproduce a whole they were never fit to, is genuinely out of sample.

Version 2 — independent synthesis from the literature review

Every additional centimeter of global mean sea-level rise (GMSLR) exposes roughly 1.6 million more people to annual coastal flooding (*Kulp and Strauss*, 2019), and unmitigated damages are projected to reach US\$14–18T (trillion) per year once rise reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018). Sea-level projections meant to guide adaptation against those costs have been published for more than four decades. A database of 756 such projections drawn from 103 studies shows their production concentrated in a handful of wealthy countries and their horizons only recently extended past 2100, and finds that, prior to the IPCC’s Sixth Assessment Report (AR6), individual studies’ high-end estimates routinely exceeded the IPCC’s own central ranges (*Garner et al.*, 2026). Over that same period, successive IPCC assessment cycles have revised their component treatments and uncertainty structure substantially (*Slangen et al.*, 2023; *Garner et al.*, 2018), evidence less of an immature field than of one still searching for the right relationship between physical detail and the observational record available to constrain it.

A growing, cross-institutional view holds that the way forward is not more physical detail but tighter observational constraint. A recent synthesis spanning the leading process-based and observational sea-level research groups argues explicitly for narrowing projections with data and for characterizing uncertainty structure, not just magnitude, in both the observations and the models (*Felikson et al.*, 2025). That argument responds to a specific, well-documented weakness of the process-based ice-sheet models that dominate current assessments, including AR6’s own 0.4–0.7 m projection for 21st-century rise (*Fox-Kemper et al.*, 2021). In hindcasting exercises, these models diverge from the historical GMSLR record (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricke et al.*, 2025), disagreements traced to structural gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025).

Component-summation frameworks, which decompose GMSLR into its physical contributors and combine them probabilistically, already offer a partial answer, and existing implementations differ mainly in what each component is calibrated against: expert-assessed probability distributions (*Kopp et al.*, 2014), independently calibrated semi-empirical response functions (*Mengel et al.*, 2016), jointly calibrated physical models (*Wong et al.*, 2017), or reduced-complexity emulators of process-model output (*Nauels et al.*, 2017), with the modular platform now underlying AR6 designed to accommodate all of these calibration strategies within a common structure (*Kopp et al.*, 2023). What none of them do is hold out the aggregate observational record, global sea level’s own sensitivity to global temperature, entirely from every component-level fit and use it solely as an independent check on the summed result.

The stakes of that calibration choice are concentrated in West Antarctica. Marine ice sheet instability does not simply widen the uncertainty on Antarctic contributions; it mathematically skews the resulting distribution toward high outcomes (*Robel et al.*, 2019), and independent regional ocean modeling finds that ocean-driven melt of West Antarctic ice shelves accelerates at similar rates across emissions scenarios this century, meaning much of that risk may already be locked in (*Naughten et al.*, 2023). Recent work also finds that the near-term rate of Antarctic mass loss is itself predictive of the rate over the following 3–5 decades (*McCormack et al.*, 2026), suggesting the tail is not only large but partially foreseeable from the current observational record. Yet deep-uncertainty treatments that represent this risk as a structured scenario mixture rather than a single process-model trajectory remain rare (e.g., *Bakker et al.*, 2017). A recent variance decomposition reaches a seemingly contrary conclusion, that emissions-scenario choice dominates projection uncertainty through the mid-century before geophysical uncertainty takes over (*Darnell et al.*, 2025), a claim that a framework treating the tail explicitly cannot simply set aside.

None of this is an argument against physical detail; it is an argument for testing that detail against data before trusting it. Outside sea-level rise specifically, climate science has long recognized that a hierarchy spanning idealized and comprehensive models serves understanding, and often forecast robustness, better than comprehensive models alone (*Held*, 2005), a pattern general forecasting research confirms empirically across some 100,000 time series (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). Semi-empirical models that regress GMSLR directly onto GMST provide another genuinely data-driven benchmark (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009), distinguishing the choice of model complexity from the choice of calibration source. An even simpler benchmark needs no temperature regression at all: a quadratic extrapolation of the satellite-observed GMSL record alone agrees with IPCC SROCC and AR6 projections to within their own stated uncertainty (*Nerem et al.*, 2022). Nor is the record uniformly unkind to assessment-based projection. An earlier IPCC assessment’s mid-range 21st-century sea-level projection verified to within about a centimeter three decades later (*Törnqvist et al.*, 2025), a reminder that any critique of process-based projections should be aimed at specific assessment cycles and specific components, not at process-based reasoning as such.

Here, we build a component-level framework that puts that discipline into practice directly: every physical contributor to GMSLR, ocean thermal expansion, Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge, West Antarctic Ice Sheet (WAIS) glacier discharge, East Antarctic Ice Sheet (EAIS) surface mass balance, the Antarctic Peninsula,

glaciers outside the ice sheets, and terrestrial water storage, is calibrated with its own parsimonious physical model against an independent observational record, using Bayesian inference to propagate parameter uncertainty rather than adopting it from assessment or process-model output. We call the result SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. The only genuinely new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response; every other component adapts an existing formulation. Its treatment of WAIS is a structured deep-uncertainty scenario mixture built to represent the skewed tail described above rather than average over it. Observed GMSLR and its GMST sensitivity, withheld from every component calibration, are reserved as the composite-level diagnostic that current frameworks do not report, and against which SPECTRUM’s aggregated components are tested here for the first time.

Version 3 — condensed synthesis

Global mean sea-level rise (GMSLR) already exposes roughly 1.6 million more people to annual flooding for every additional centimeter (*Kulp and Strauss*, 2019), with unmitigated damages projected at US\$14–18T (trillion) per year once rise reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018). Sea-level projections meant to inform that risk have been published for four decades and span a wide range of methods and outcomes (*Garner et al.*, 2026). Rates of GMSLR have already doubled twice since the mid-20th century (*Frederikse et al.*, 2020; *Hamlington et al.*, 2024), yet coastal planning still leans heavily on the IPCC Sixth Assessment Report (AR6), which projects 0.4–0.7 m of 21st-century rise (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). A growing, cross-institutional view holds that narrowing AR6’s range requires tighter observational constraint on projections, not more physical detail (*Felikson et al.*, 2025).

Three model classes have taken up that challenge, each with a real weakness. Process-based ice-sheet models resolve ice flow and fracture directly, but they demand more constraints than the observational record can currently supply. In hindcasting studies, they diverge from the historical GMSLR record (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). These disagreements trace to gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025). Decades of general forecasting research show that models with more free parameters than their data can constrain tend to forecast worse than simpler alternatives (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). Semi-empirical models that regress the rate of GMSLR onto global mean surface temperature are simple and transparent, but the underlying relationship is not obviously physical, causal, or stationary (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Church et al.*, 2013). Even so, such fits track the observed record at least as well as contemporaneous process-based projections (*Rahmstorf et al.*, 2012). An even simpler benchmark needs no temperature regression at all: a quadratic extrapolation of the satellite-observed GMSL record alone agrees with IPCC SROCC and AR6 projections to within their own stated uncertainty (*Nerem et al.*, 2022). Between these extremes, component-level models decompose GMSLR into its physical contributors and calibrate each against existing data (*Kopp et al.*,

2014; Mengel *et al.*, 2016; Wong *et al.*, 2017), trading basin-scale resolution for parsimony and a direct physical connection, at the cost of assuming stationarity. None of the published implementations, however, withholds the aggregate observational record from every component fit, so none can use it as an independent check on the summed result. Ordering these three model classes by complexity, rather than treating any one as authoritative, turns their disagreement into a diagnostic, letting the simplest, most transparent models motivate and check the more complex ones.

The stakes of this missing independent check are concentrated in West Antarctica, where marine ice sheet instability does not merely widen projection uncertainty but skews it toward high outcomes (Robel *et al.*, 2019), and where independent regional ocean modeling finds melt accelerating at similar rates across emissions scenarios this century, meaning much of that risk may already be locked in (Naughten *et al.*, 2023). This West Antarctic risk differs structurally from the temperature-sensitive uncertainty in the other components. Treating it as a single central trajectory, rather than a mixture spanning its full range, would understate exactly the high-end damages introduced above.

Here, we build a component-level framework in which each contributor to GMSLR (ocean thermal expansion; Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge; West Antarctic Ice Sheet (WAIS) glacier discharge; East Antarctic Ice Sheet (EAIS) surface mass balance; the Antarctic Peninsula; glaciers outside the ice sheets; and terrestrial water storage) gets its own parsimonious physical model, calibrated against an independent observational record using Bayesian inference. We call it SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Its only new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response. Its treatment of WAIS is a structured deep-uncertainty scenario mixture, built to represent the skewed tail described above rather than average over it. Observed GMSLR and its sensitivity to GMST, the relationship at the core of the semi-empirical class described above, are withheld from every component calibration and used only as a diagnostic on the composite model. The central test is therefore out of sample. It asks whether the independently calibrated parts reproduce a whole they were never fit to.

Version 4 — critically edited from Version 3 (draft A): thesis-first, four-line WAIS case

Global mean sea-level rise (GMSLR) already exposes roughly 1.6 million more people to annual flooding for every additional centimeter (Kulp and Strauss, 2019), with unmitigated damages projected at US\$14–18T (trillion) per year once rise reaches 0.9–1.1 m (Jevrejeva *et al.*, 2018). Sea-level projections meant to inform that risk have been published for four decades and span a wide range of methods and outcomes (Garner *et al.*, 2026). Rates of GMSLR have already doubled twice since the mid-20th century (Frederikse *et al.*, 2020; Hamlington *et al.*, 2024). Coastal planning nonetheless leans heavily on the IPCC Sixth Assessment Report (AR6), which projects 0.4–0.7 m of 21st-century rise (Fox-Kemper *et al.*, 2021; Hirschfeld *et al.*, 2023). A growing, cross-institutional view holds that narrowing AR6’s range requires tighter observational constraint on projections, not more physical detail (Felikson *et al.*, 2025).

Three model classes have taken up that challenge, each with a real weakness. Process-based ice-sheet models resolve ice flow and fracture directly, but they demand more constraints than the observational record can currently supply. In hindcasting studies, they diverge from the historical GMSLR record (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). These disagreements trace to gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025). Decades of general forecasting research show that models tend to forecast worse than simpler alternatives once they carry more free parameters than their data can constrain (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). Semi-empirical models regress the rate of GMSLR onto global mean surface temperature. They are simple and transparent, but the underlying relationship is not obviously physical, causal, or stationary (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Church et al.*, 2013). Even so, such fits track the observed record at least as well as contemporaneous process-based projections (*Rahmstorf et al.*, 2012). An even simpler benchmark needs no temperature regression at all: a quadratic extrapolation of the satellite-observed GMSL record alone agrees with IPCC SROCC and AR6 projections to within their own stated uncertainty (*Nerem et al.*, 2022). Between these extremes, component-level models decompose GMSLR into its physical contributors and calibrate each against existing data (*Kopp et al.*, 2014; *Mengel et al.*, 2016; *Wong et al.*, 2017), trading basin-scale resolution for parsimony and a direct physical connection, while assuming the relationships they calibrate are stationary. None of the published implementations, however, withholds the aggregate observational record from every component fit, so none can use it as an independent check on the summed result. Rather than treat any one model class as authoritative, we order the three by complexity and treat their disagreement as a diagnostic. The simplest, most transparent models motivate and check the more complex ones.

The stakes of this missing independent check are concentrated in West Antarctica. Four independent lines of evidence converge on the same conclusion. West Antarctica is the largest source of uncertainty in sea-level projections, and its share of that uncertainty is growing.

First, the observational record shows the loss is already under way. Antarctica’s grounding lines have retreated for thirty consecutive years (*Rignot et al.*, 2026), and the Thwaites Glacier basin in West Antarctica may already be undergoing an unstable, self-sustaining collapse (*Joughin et al.*, 2014), with recent observations there consistent with continued high rates of loss over the next fifty years (*Goldberg et al.*, 2026). Second, a structured survey of expert judgment, built independently of any process-model ensemble, assigns West Antarctica wider and more heavily tailed contribution distributions than those ensembles do (*Bamber et al.*, 2019). Third, that spread is not merely disagreement between models talking past each other. A single ice-sheet model’s own uncertainty in calving, basal sliding, ice-shelf rheology, and ocean forcing, propagated forward for centuries, already produces a wide range of Antarctic outcomes (*Bulthuis et al.*, 2019), and the largest multi-model ensemble, ISMIP6, adds still more disagreement among its Antarctic members than among the members representing any other component (*Seroussi et al.*, 2020). Fourth, that disagreement is not fixed in time. A variance decomposition of projected sea-level rise finds emissions-scenario choice dominant through mid-century, after which geophysical uncertainty, chiefly Antarctic ice dynamics, becomes the largest term (*Darnell et al.*, 2025), and most of the emissions-pathway spread in total projections traces to the Antarctic contribution alone (*DeConto*

et al., 2021). One recent finding cuts the other way: the near-term rate of Antarctic mass loss appears to carry predictive information about the rate over the following three to five decades (*McCormack et al.*, 2026).

The physical mechanism behind West Antarctic loss argues against reading that predictive signal as reassurance. Marine ice sheet instability does not merely widen projection uncertainty. It skews the distribution toward high outcomes (*Robel et al.*, 2019). Coupled with hydrofracturing and ice-cliff collapse, that instability could add more than a meter of Antarctic sea-level rise by 2100 under unabated emissions (*DeConto and Pollard*, 2016). Independent regional ocean modeling finds shelf melt accelerating at similar rates across emissions scenarios this century, meaning much of that risk is already locked in regardless of future mitigation (*Naughten et al.*, 2023). West Antarctic risk therefore differs structurally from the temperature-sensitive uncertainty in the other components. It is skewed, partly irreducible, and growing in relative importance as the century proceeds, and even the choice of which physical processes to include, not just which parameters to assume, changes the shape of its probability distribution (*Kopp et al.*, 2017). Some published projections already treat it that way, as a deep-uncertainty mixture of scenarios rather than a single central trajectory (*Bakker et al.*, 2017). Anything less would understate exactly the high-end damages introduced above.

Here, we build a component-level framework that separates GMSLR into seven contributors: ocean thermal expansion; Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge; West Antarctic Ice Sheet (WAIS) glacier discharge; East Antarctic Ice Sheet (EAIS) surface mass balance; the Antarctic Peninsula; glaciers outside the ice sheets; and terrestrial water storage. Each contributor gets its own parsimonious physical model, calibrated against an independent observational record using Bayesian inference. We call it SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Its only new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response. Its treatment of WAIS is a structured deep-uncertainty scenario mixture, built to represent the skewed tail described above rather than average over it. We withhold observed GMSLR and its sensitivity to GMST, the relationship at the core of the semi-empirical class described above, from every component calibration, and use it only as a diagnostic on the composite model. The central test is therefore out of sample. It asks whether the independently calibrated parts reproduce a whole they were never fit to.

Version 5 — critically edited from Version 3 (draft B): three-movement WAIS argument with counter-evidence

Global mean sea-level rise (GMSLR) already exposes roughly 1.6 million more people to annual flooding for every additional centimeter (*Kulp and Strauss*, 2019), with unmitigated damages projected at US\$14–18 trillion per year once rise reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018). Projections of that risk span four decades and a wide range of methods and outcomes (*Garner et al.*, 2026). Rates of GMSLR have already doubled twice since the mid-20th century (*Frederikse et al.*, 2020; *Hamlington et al.*, 2024), yet coastal planning still leans heavily on the IPCC Sixth Assessment Report (AR6), which projects 0.4–0.7 m

of 21st-century rise (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). A growing, cross-institutional view holds that narrowing AR6’s range requires tighter observational constraint on projections, not more physical detail (*Feliksion et al.*, 2025).

Three model classes have taken up that challenge, each with a real weakness. Process-based ice-sheet models resolve ice flow and fracture directly, but they demand more constraints than the observational record can currently supply. In hindcasting studies, they diverge from the historical GMSLR record (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). These disagreements trace to gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025). Forecasting research spanning decades and many fields finds the same pattern. Models with more free parameters than their data can constrain tend to forecast worse than simpler alternatives (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). Semi-empirical models that regress the rate of GMSLR onto global mean surface temperature are simple and transparent, but the underlying relationship is not obviously physical, causal, or stationary (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Church et al.*, 2013). Even so, such fits track the observed record at least as well as contemporaneous process-based projections (*Rahmstorf et al.*, 2012). An even simpler benchmark needs no temperature regression at all: a quadratic extrapolation of the satellite-observed GMSL record alone agrees with IPCC SROCC and AR6 projections to within their own stated uncertainty (*Nerem et al.*, 2022). Between these extremes sit component-level models, which decompose GMSLR into its physical contributors and calibrate each against existing data (*Kopp et al.*, 2014; *Mengel et al.*, 2016; *Wong et al.*, 2017). This approach trades basin-scale resolution for parsimony and a direct physical connection, at the cost of assuming stationarity. None of the published implementations, however, withholds the aggregate observational record from every component fit, so none can check the summed result against it independently. No one class is authoritative on its own. Ranked instead by complexity, the three become a diagnostic. The simplest, most transparent models motivate and check the more complex ones.

The stakes of that missing check are concentrated in West Antarctica. Marine ice sheet instability there does not merely widen the uncertainty in sea-level projections, it skews that uncertainty toward high outcomes (*Robel et al.*, 2019). A second, compounding process operates there too. Hydrofracturing strips away the ice shelves that buttress tall ice cliffs, letting them collapse under their own weight. The model that first identified this mechanism found it could add more than a meter to Antarctica’s contribution by 2100 (*DeConto and Pollard*, 2016). How much of that process will actually occur remains itself disputed. Recalibrating the same physics against observations from the last interglacial, the Pliocene, and the past 25 years found no need to invoke ice-cliff collapse at all, and put the resulting 95th-percentile Antarctic contribution below 43 cm this century (*Edwards et al.*, 2019). West Antarctica therefore carries a layer of uncertainty the other components do not. Two things are unresolved there, not one: whether an instability occurs at all, and how large it would be if it did.

Thwaites Glacier, the drainage basin most likely to host either instability, may already be undergoing a self-sustaining retreat that cannot be stopped, according to observations and modeling published as early as 2014 (*Joughin et al.*, 2014). Later radar surveys found that its retreat proceeds unevenly along the grounding line, with warm water undercutting some

sectors ten times faster than others (*Milillo et al.*, 2019). More recent observations remain consistent with continued high rates of loss there over the next 50 years (*Goldberg et al.*, 2026), and grounding lines across Antarctica have retreated persistently for three decades (*Rignot et al.*, 2026). Independent regional ocean modeling adds a further finding. Ice-shelf melt accelerates at similar rates whether emissions are cut sharply or not this century, so much of this risk is already locked in regardless of the mitigation path taken (*Naughten et al.*, 2023).

Several other, methodologically independent assessments converge on the same conclusion: West Antarctica dominates the uncertainty budget for sea-level rise. Two independent multi-model exercises find deep disagreement in Antarctica’s own projected contribution. ISMIP6’s ensemble of process models disagrees even on its sign, spanning outcomes from a small sea-level fall to 30 cm of rise by 2100 under high emissions (*Seroussi et al.*, 2020), while a sixteen-model synthesis built on linear response theory finds a comparably wide spread in basal-melt-driven mass loss (*Levermann et al.*, 2020). A structured survey of expert glaciological judgment, drawing on physical reasoning rather than model runs, arrives independently at similarly wide, high-tailed distributions (*Bamber et al.*, 2019). At the emissions-scenario level, Antarctica’s own total contribution swings more dramatically with how much warming is avoided than any other component does (*DeConto et al.*, 2021). A decomposition of total sea-level-rise variance shows exactly when the balance tips. Emissions-scenario choice dominates uncertainty through mid-century, after which geophysical uncertainty, chiefly Antarctic ice dynamics, becomes the largest term (*Darnell et al.*, 2025). Under high emissions, the IPCC’s own low-likelihood, high-impact storyline for 2100 combines earlier-than-projected ice-shelf disintegration, abrupt onset of these Antarctic instabilities, and faster Greenland change to add more than a meter of additional rise (*Fox-Kemper et al.*, 2021). The near-term rate of Antarctic mass loss even appears to predict its own rate three to five decades ahead (*McCormack et al.*, 2026), a relationship too newly reported to narrow this range yet. This risk is structurally different from the temperature-sensitive uncertainty in the other components. A single central WAIS trajectory, rather than a mixture spanning its full range, would understate exactly the high-end damages described at the outset.

Here we build a component-level framework that gives each contributor to GMSLR its own parsimonious physical model, calibrated against an independent observational record using Bayesian inference. The contributors are ocean thermal expansion; Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge; West Antarctic Ice Sheet (WAIS) glacier discharge; East Antarctic Ice Sheet (EAIS) surface mass balance; the Antarctic Peninsula; glaciers outside the ice sheets; and terrestrial water storage. We call it SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Its only new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response. Its treatment of WAIS is a structured deep-uncertainty scenario mixture (*Bakker et al.*, 2017), built to represent the skewed tail described above rather than average over it. We withhold observed GMSLR and its sensitivity to GMST, the relationship at the core of the semi-empirical class described above, from every component calibration, and use it only as a diagnostic on the composite model. The central test is therefore out of sample. It asks whether the independently calibrated parts reproduce a whole they were never fit to.

Version 6 — merged synthesis of Versions 3–5, WAIS transition corrected

Global mean sea-level rise (GMSLR) already exposes roughly 1.6 million more people to annual flooding for every additional centimeter (*Kulp and Strauss*, 2019), with unmitigated damages projected at US\$14–18T (trillion) per year once rise reaches 0.9–1.1 m (*Jevrejeva et al.*, 2018). Sea-level projections meant to inform that risk have been published for four decades and span a wide range of methods and outcomes (*Garner et al.*, 2026). Rates of GMSLR have already doubled twice since the mid-20th century (*Frederikse et al.*, 2020; *Hamlington et al.*, 2024). Coastal planning nonetheless leans heavily on the IPCC Sixth Assessment Report (AR6), which projects 0.4–0.7 m of 21st-century rise (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). A growing, cross-institutional view holds that narrowing AR6’s range requires tighter observational constraint on projections, not more physical detail (*Felikson et al.*, 2025).

Three model classes have taken up that challenge, each with a real weakness. Process-based ice-sheet models resolve ice flow and fracture directly, but they demand more constraints than the observational record can currently supply. In hindcasting studies, they diverge from the historical GMSLR record (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). These disagreements trace to gaps such as an assumed ice viscosity far less sensitive to stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025). Forecasting research spanning decades and many fields finds the same pattern. Models with more free parameters than their data can constrain tend to forecast worse than simpler alternatives (*Green and Armstrong*, 2015; *Makridakis et al.*, 2020). Semi-empirical models regress the rate of GMSLR onto global mean surface temperature. They are simple and transparent, but the underlying relationship is not obviously physical, causal, or stationary (*Rahmstorf*, 2007; *Vermeer and Rahmstorf*, 2009; *Church et al.*, 2013). Even so, such fits track the observed record at least as well as contemporaneous process-based projections (*Rahmstorf et al.*, 2012). An even simpler benchmark needs no temperature regression at all: a quadratic extrapolation of the satellite-observed GMSL record alone agrees with IPCC SROCC and AR6 projections to within their own stated uncertainty (*Nerem et al.*, 2022). Between these extremes, component-level models decompose GMSLR into its physical contributors and calibrate each against existing data (*Kopp et al.*, 2014; *Mengel et al.*, 2016; *Wong et al.*, 2017), trading basin-scale resolution for parsimony and a direct physical connection, while assuming the relationships they calibrate are stationary. None of the published implementations, however, withholds the aggregate observational record from every component fit, so none can use it as an independent check on the summed result. No one class is authoritative on its own. Ranked instead by complexity, the three become a diagnostic. The simplest, most transparent models motivate and check the more complex ones.

West Antarctica is the largest single source of uncertainty in sea-level projections. That is not a consequence of how any model class chooses to calibrate; it follows from the ice physics and the observational record themselves. Four independent lines of evidence converge on this conclusion, and West Antarctica’s share of the total uncertainty is growing.

First, the observational record shows the loss is already under way and proceeding un-

evenly. Antarctica’s grounding lines have retreated for thirty consecutive years (*Rignot et al.*, 2026), and the Thwaites Glacier basin may already be undergoing a self-sustaining collapse (*Joughin et al.*, 2014). Radar surveys there find warm water undercutting some sectors of the grounding line ten times faster than others (*Milillo et al.*, 2019), and recent observations remain consistent with continued high rates of loss over the next fifty years (*Goldberg et al.*, 2026). Second, a structured survey of expert judgment, built independently of any process-model ensemble, assigns West Antarctica wider and more heavily tailed contribution distributions than those ensembles do (*Bamber et al.*, 2019). Third, that spread is not merely disagreement between models talking past each other. A single ice-sheet model’s own uncertainty in calving, basal sliding, ice-shelf rheology, and ocean forcing, propagated forward for centuries, already produces a wide range of Antarctic outcomes (*Bulthuis et al.*, 2019). The largest multi-model ensemble, ISMIP6, disagrees even on the sign of Antarctica’s contribution (*Seroussi et al.*, 2020), and a sixteen-model synthesis built on linear response theory finds a comparably wide spread in basal-melt-driven mass loss (*Levermann et al.*, 2020). Fourth, that disagreement is not fixed in time. A variance decomposition of projected sea-level rise finds emissions-scenario choice dominant through mid-century, after which geophysical uncertainty, chiefly Antarctic ice dynamics, becomes the largest term (*Darnell et al.*, 2025), and most of the emissions-pathway spread in total projections traces to the Antarctic contribution alone (*DeConto et al.*, 2021). Under high emissions, the IPCC’s own low-likelihood, high-impact storyline for 2100 combines earlier-than-projected ice-shelf disintegration, abrupt onset of marine ice sheet instability, and faster Greenland change to add more than a meter of additional rise (*Fox-Kemper et al.*, 2021). One recent finding cuts the other way: the near-term rate of Antarctic mass loss appears to carry predictive information about the rate over the following three to five decades (*McCormack et al.*, 2026), though this relationship is too newly reported to narrow the range yet.

The physical mechanism behind West Antarctic loss argues against reading that predictive signal as reassurance. Marine ice sheet instability does not merely widen projection uncertainty; it skews the distribution toward high outcomes (*Robel et al.*, 2019). Coupled with hydrofracturing and ice-cliff collapse, that instability could add more than a meter of Antarctic sea-level rise by 2100 under unabated emissions (*DeConto and Pollard*, 2016). How much of that process will actually occur is itself disputed: recalibrating the same physics against observations from the last interglacial, the Pliocene, and the past 25 years finds no need to invoke ice-cliff collapse at all, bounding the 95th-percentile Antarctic contribution below 43 cm this century without it (*Edwards et al.*, 2019). Which physical processes to assume, not only which parameters to fit, therefore changes the shape of West Antarctica’s projected distribution (*Kopp et al.*, 2017). Independent regional ocean modeling adds a further constraint: shelf melt accelerates at similar rates across emissions scenarios this century, so much of this risk is already locked in regardless of the mitigation path taken (*Naughten et al.*, 2023). This uncertainty is structurally different from the temperature-sensitive uncertainty in the other components, which is why some published projections already treat West Antarctica as a deep-uncertainty mixture of scenarios rather than a single central trajectory (*Bakker et al.*, 2017). Anything less would understate exactly the high-end damages introduced above.

No published component-level framework performs the independent check described above, and the gap matters most exactly where the stakes are largest: West Antarctica. Here, we

build a framework in which each contributor to GMSLR gets its own parsimonious physical model, calibrated against an independent observational record using Bayesian inference. The contributors are ocean thermal expansion; Greenland Ice Sheet (GrIS) surface mass balance and glacier discharge; West Antarctic Ice Sheet (WAIS) glacier discharge; East Antarctic Ice Sheet (EAIS) surface mass balance; the Antarctic Peninsula; glaciers outside the ice sheets; and terrestrial water storage. We call it SPECTRUM, for Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models. Its only new component model links GrIS discharge to ocean temperature through a 3-parameter time-lagged response. Its treatment of WAIS is a structured deep-uncertainty scenario mixture, built to represent the skewed tail described above rather than average over it. We withhold observed GMSLR and its sensitivity to GMST, the relationship at the core of the semi-empirical class described above, from every component calibration, and use it only as a diagnostic on the composite model. The central test is therefore out of sample: it asks whether the independently calibrated parts reproduce a whole they were never fit to.

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